

Breadth First Search (BFS)

Aim:

To traverse a graph using BFS.

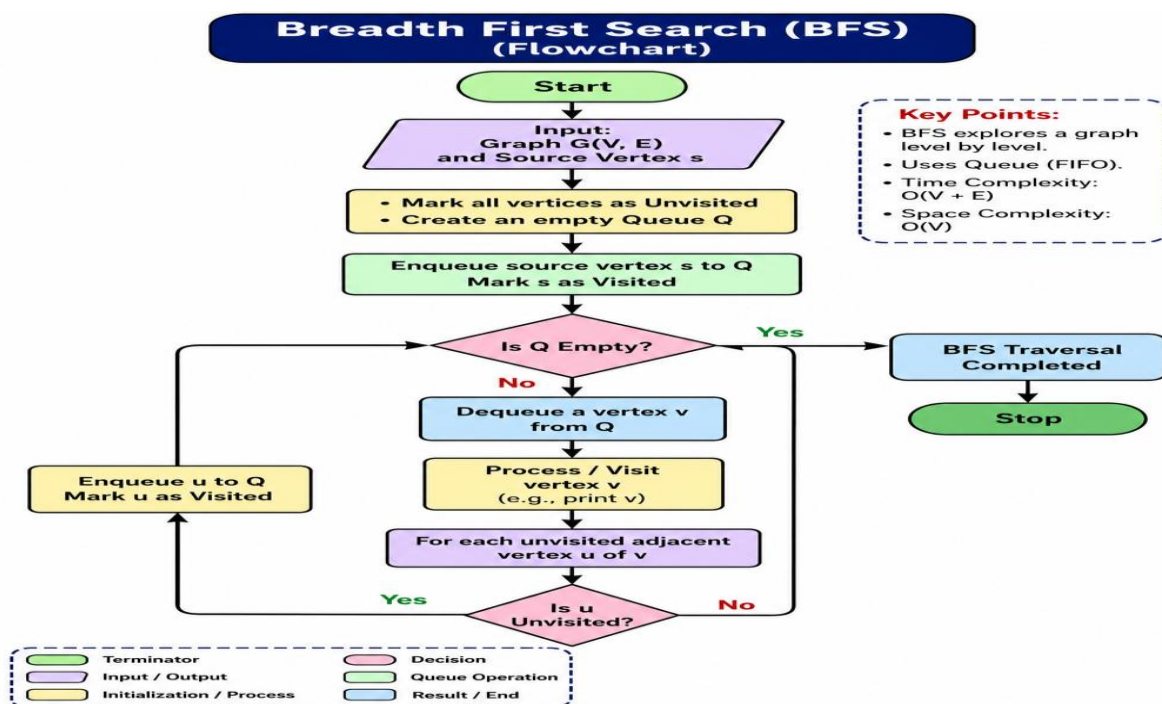
Objective:

To visit nodes level by level.

Algorithm:

1. Start from source node.
2. Add node to queue.
3. Visit adjacent nodes.
4. Repeat until queue becomes empty.

Flowchart:



Python code:

```
graph = {}
```

```
n = int(input("Enter number of nodes: "))
```

```
for i in range(n):
```

```
    node = input("Enter node name: ")
```

```
    neighbors = input(f"Enter neighbors of {node} (space-separated): ").split()
```

```
    graph[node] = neighbors
```

```
start = input("Enter starting node: ")
```

```
visited = []
```

```
queue = [start]
```

```
while queue:
```

```
    node = queue.pop(0)
```

```
    if node not in visited:
```

```
        visited.append(node)
```

```
        queue.extend(graph[node])
```

```
print("BFS Traversal:", visited)
```

output:

```
IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Sheik Mohamed/bfs.py =====
Enter number of nodes: 5
Enter node name: a
Enter neighbors of a (space-separated): b c
Enter node name: b
Enter neighbors of b (space-separated): c d
Enter node name: c
Enter neighbors of c (space-separated): d
Enter node name: d
Enter neighbors of d (space-separated):
Enter node name: e
Enter neighbors of e (space-separated):
Enter starting node: a
BFS Traversal: ['a', 'b', 'c', 'd']
>>>
```

Result:

Graph traversed successfully.

Conclusion:

BFS visits nodes level by level.